



RUPAN CHAKKARAVARTHY E

SENIOR ENGINEER

PERSONAL PROFILE

Senior Engineer with 9+ years of expertise in VR/AR simulation, defense simulation systems, and immersive interactive applications. Proven track record in leading teams, developing fighter jet simulators (Unity3D, FlightGear), and integrating XR hardware (Varjo, Meta Quest, Leap Motion). Strong skills in Unity 3D, C#, C++, C, Qt, ROS2 OpenGL, and real-time systems, with experience in embedded programming (ADA), shader development, and AI integration. Successfully delivered projects for Aerospace, Defence . museums, and gaming. Passionate about cutting-edge XR solutions and cross-platform development.

WORK EXPERIENCE

Project Manager

MicroGenesis Techsoft Pvt LTD

Mar 2024 - Present

- Deployed in Aeronautical Development Agency (a Defence Research and Development Organisation, Ministry of Defence, India) for the Development of Indigenous Extended Reality Realtime Pilot Ergonomic Study During Simulation
- Deployed in Center for Artificial Intelligence and Robotics (CAIR-DRDO) as a Technical Manager for GIS, Realtime and Robotics Projects.
- Work Closely with Senior Scientists to gather requirements and feedback.
- Conduct day-to-day project coordination, planning, and implementation with the team
- Create functional and technical application documents

Freelance Software Engineer

Mar 2023 - Mar 2024

- Development of Interactive Applications
- Conduct day-to-day project coordination, planning, and implementation with the team
- Create functional and technical application documents

CONTACT

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SKILLS

C#
C++
C
Qt
ROS2
NAV2
OpenGL
Unity 3D
OpenCV
FlightGear
Open VR
XLIB
XR Interaction Toolkit
ARCore
ARFoundation
Vuforia
SQL
Node JS
PHP

Steam VR

Web Development

Oculus Integration

ADA - Embedded
Programming
Language

One Test Embedded
PTU Testcase
Generation

Substance Painter

Blender

Project Management

Senior Engineer
COMAVIA Systems Technologies Pvt LTD
Nov 2021 - Feb 2023

Senior Software Engineer (XR)
BAeHAL Software Limited
Nov 2020 - Oct 2021
Jan 2019 - October 2019

- Deployed in Aeronautical Development Agency (a Defence Research and Development Organisation, Ministry of Defence, India) for the Development of Indigenous Virtual Reality Fighter Jet Simulator
- Work Closely with Senior Scientists to gather requirements and feedback.
- Lead the Team to Successfully Deploy the Simulator as per requirements
- Conduct day-to-day project coordination, planning, and implementation with the team
- Create functional and technical application documents

Senior XR Engineer
D2VR Technologies | Oct 2019 - Oct 2020

- Development of VR and AR applications, prototypes and new functionalities
- Agile development planning
- Running tests and code reviews
- Identification of risks and opportunities during development
- Identify opportunities to improve general workflows
- Write and maintain software system documentation

XR Engineer
D2VR Technologies | July 2018 - Dec 2018

- Development of VR and AR applications
- Create and Improve VR Interactions
- Debugging and Documentation
- Creating Prototypes for XR use cases

Unity 3D Developer
Satya Aerospace Tech Data Services Pvt. Ltd
05/2016 – 06/2018

- Worked as Game Programmer and at times as animator and Game Environment Integrator.
- Network Programming for the gameplay
- Worked Hands on AR SDK's Such as Vuforia , ARcore OpenVR and Google VR SDK's.
- Build VR Applications for Windows Mixed Reality , Oculus RIFT and HTC Vive using OpenVR SDK and Steam VR Plugin.

DEVICE EXPERTISE

Varjo XR 3 and 4

Meta Quest Series

Oculus Rift

HTC Vive Pro

Windows MR Headsets

Leap Motion

Android Devices

iPad

Sense Glove - Force
and Haptic Feedback
Gloves

XSENS Movella Motion
Capture Suit

KEY PROJECTS AND ACHIEVEMENTS

- **Cockpit Display System (CDS) Development for Fighter Aircrafts (MK2 , AMCA) (OpenGL , C++ , C)**
 - Developed the Cockpit Display System (CDS) compliant with ARINC 661 Standard.
 - Utilized C and OpenGL for low-latency rendering of complex SVG-based display pages.
 - Implemented shader-based rendering to handle real-time graphical elements efficiently.
 - Designed multi-partition dynamic displays with support for modular updates.
 - Created a custom Menu Module for user interaction and control.
 - Integrated Epiphan-based video capture system with real-time cropping capability.
 - Enabled video streaming and screen capture functionalities using Python/C++ and OpenCV.
 - Team Lead
- **Extended Reality Indigenous Fighter Jet Simulator (Unity 3D)**
 - Development of multiple systems with respect to the simulator
 - AR Visualization of Cockpit Controls Interaction using AR-Foundation.
 - Development of Custom Shaders manipulating world space UV for HUD Simulation.
 - Development of physics-based animations using joints for cockpit switches and indicators.
 - Automation of scripts and component attachment to cockpit controls derived from excel as per type and position using Unity Editor Scripting.
 - Integration of Leap Motion device for cockpit interaction using Hand Tracking
 - Development of Environment with Day/Night, Weather simulation using Enviro Plugin.
 - Development of 6 DOF Hand Tracked Visualization of Cockpit using Oculus Quest 2
 - Development and Scheduling of Symbolology Drawing for Multiple Cockpit Display Surfaces
 - Team Lead
- **Real-time Pilot Ergonomic Analysis(Unity 3D)**
 - Motion capture Integration from Xsens Movella
 - Developed a Bone Data Extraction module to enable ergonomic metric evaluation and stress analysis during high-load aircraft maneuvering.
 - Development of Data Logging and Playback Module for logging movement and posture data with timestamps, enables playback of recorded sessions for post-analysis, and supports exporting data in CSV or JSON formats for use with external tools such as MATLAB or Python.
 - Development of Report Generation which summarizes ergonomic scores, and auto-generates PDF reports for evaluation

- **ECFM LRU**
 - Test case Generation and Component Testing for Real time Airworthy unit.
 - Integration ADA Code with C++ Windows Code thru DLL and Network protocols
- **Synthetic Terrain**
 - Shader development for Synthetic Terrain generation in Flight Gear.
- **Development of Multi-Region Display in Large area Display using Windows C++ with interchangeable pages with touch input.**
- **Lane Detection python Algorithm integration with Flight gear(C,C++).**
- **Implement Video Streaming Algorithm to send and receive screen/window captured data thru UDP Protocols by encoding and decoding thru opencv and c++.**
- **Fighter Jet Sensors Simulation with Multi-Monitor & Multi-Window using FlightGear and XLIB in UBUNTU**
- **Bomb Drop Trajectory Simulation in FlightGear and Unity as per trajectory Data**
- **Maritime Situational Awareness System**
 - GIS Based Situational awareness system using Qt C++ and GIS Libraries
- **Autonomous Underwater Vehicle(ECA - A9) Software based on ROS2 Nav2 Software Stack.**
- **Deployed a Self-hosted AI backend using open-source tools for scalable, multimodal LLM management and communication**
- **Deployed Open-source AI chat interface (like ChatGPT) via Docker, integrated with a local LLM backend. Includes features like User management and Model access management.**
- **Setting up of open-source AI agents plugins in VS Code to integrate with LLMs for automated code generation, AI pair programming, smart autocompletion, and in-IDE chat assistance enabling developer efficiency and boosting productivity.**

- Self-hosted, low-latency video conferencing solution (Google Meet alternative) deployed via Docker for real-time communication.
- LIDAR and Point Cloud Data VR Visualization and Walkthrough
 - Worked Closely with MIT (Madras Institute of Technology) to bring Drone captured Lidar point Cloud data to realtime visualization and Create walkthrough and navigation systems
- MMORPG PC VR Game Development
 - Worked with a Canadian PHD Professor in Developing a MMORPG Game
 - Developed Building Systems , Procedural Terrain Generation system and Networking Modules.
- Freelance Projects
 - Development of Puli-Meka (Aadu-Puli) Strategic turn based Multiplayer game with AI for Keezhadi Archeological museum.
 - Development of Interactive Content 4 split Multi-User Touch Table For Museums
 - VR - 360 Degree Immersive tour through the museum, with audio and key annotations
- Personal Projects
 - Development of AR Sketch Application with SLAM for Anchoring Drawings in real-world space.
 - Link - <https://www.youtube.com/watch?v=OKxEMtbeyNI>

EDUCATIONAL HISTORY

Secondary Education

Vel's Vidyashram
03/2010

9.0 GPA

Higher Secondary

Vel's Higher Secondary School
03/2012

77%

Bachelor of Engineering (Computer Science)

MNM Jain Engineering College
07/2012 - 04 / 2016

6.8 GPA

- Developed an Online TollGate Payment System (TollPay) - using Phone Gap as a final year project
- Website Developer for Symposium and Cultural.